Time Series Analysis

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Contents of Lecture

- Functional/stochastic model identification
- Mathematical and statistical background
- Discrete Fourier Transform
- Univariate/multivariate model

Functional model identification

Deterministic effects in GNSS time series are:

- Functional trend is an important element for prediction
- Trend analysis and estimation
- Outlier detection, identification, and adaptation (DIA)
- Offset identification and adaptation
- Periodic signals identification and adaptation

Stochastic model identification

Problems on stochastic effects in GNSS time series are:

- Types of noise of time series
- Tools to identify appropriate noise model
- Tools to estimates amplitudes of noise components
- What can be affected by an appropriate noise model?
- Stochastic model is an important issue for prediction

Functional model identification

Mathematical tools:

- Application of least squares principle
- Applications of hypotheses testing

A few difficulties:

- Presence of gaps
- Unknown structure of noise
- Priorities on stochastic and functional models

Mathematical/Statistical Background

Principle of least squares:

$$E(\underline{y}) = Ax \text{ and } D(\underline{y}) = Q_{yy}$$
$$y = Ax + e$$

Then the weighted least-squares solution of the system is defined as

$$\hat{x} = \arg \min_{x \in \mathbb{R}^n} (y - Ax)^T W (y - Ax)$$

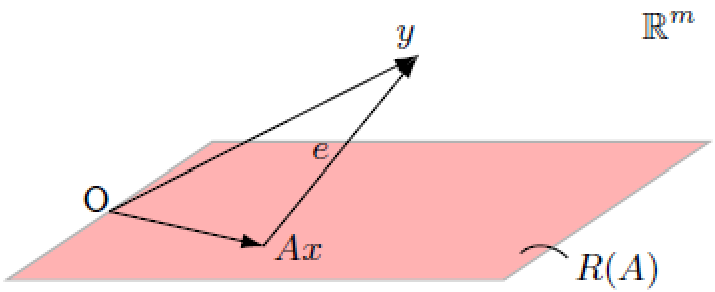
The weighted least-squares solution can be computed from y, A and W as

$$\begin{cases} \underline{\hat{x}} & = & (A^T W A)^{-1} A^T W \underline{y} \\ \underline{\hat{y}} & = & P_A \underline{y} \\ \underline{\hat{e}} & = & P_A^{\perp} \underline{y} \end{cases}$$

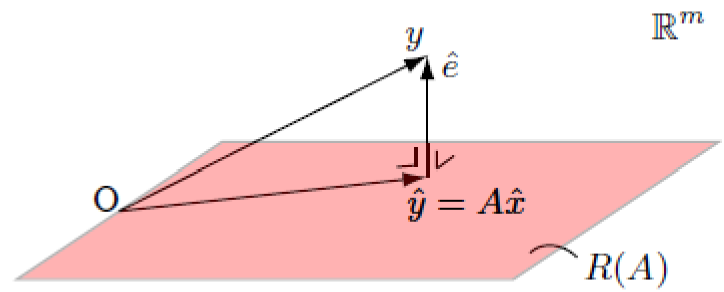
$W = Q_{yy}^{-1}$: Best linear unbiased estimation (BLUE)

$$Q_{\hat{x}\hat{x}} = (A^T Q_{yy}^{-1} A)^{-1}, \quad Q_{\hat{y}\hat{y}} = A(A^T Q_{yy}^{-1} A)^{-1} A^T, \quad Q_{\hat{e}\hat{e}} = Q_{yy} - Q_{\hat{y}\hat{y}}$$

Mathematical/Statistical Background



(a)



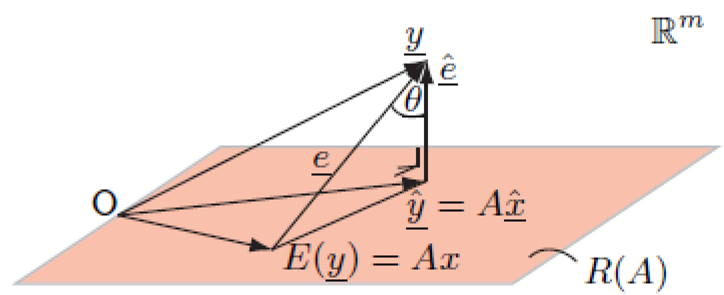
(b)

$$\|y\|_W^2 \quad = \quad \|\hat{y}\|_W^2 + \|\hat{e}\|_W^2 \quad \text{(Pythagoras)} \qquad\qquad W = Q_{yy}^{-1}$$

$$\|\underline{y} - Ax\|^2_{Q_{yy}^{-1}} = \|\underline{\hat{y}} - Ax\|^2_{Q_{yy}^{-1}} + \|\underline{\hat{e}}\|^2_{Q_{yy}^{-1}} \sim \chi^2(m, 0)$$

$$\|\underline{\hat{y}} - Ax\|^2_{Q_{yy}^{-1}} = \|\underline{\hat{x}} - x\|^2_{Q_{\hat{x}\hat{x}}^{-1}} \sim \chi^2(n, 0)$$

$$\|\underline{\hat{e}}\|^2_{Q_{yy}^{-1}} = \|\underline{t}\|^2_{Q_{tt}^{-1}} \sim \chi^2(m - n, 0)$$



Mathematical/Statistical Background

Hypothesis testing in a linear model:

The m-vector of observables $\underline{y}$ is assumed to be normally distributed

$$\underline{y} \sim \mathrm{N}_m(E\left(\underline{y}\right), Q_{yy})$$

The following two composite hypotheses on the expectation of $\underline{y}$ are put forward

$$H_o:\ E\left(\underline{y}\right) = Ax$$

$$H_a:\ E\left(\underline{y}\right) = Ax + C_y\nabla$$

Either accepts or rejects null hypothesis

Mathematical/Statistical Background

Null hypothesis is rejected if:

$$\text{reject } H_o \text{ if } \hat{e}_o^T Q_{yy}^{-1} \hat{e}_o - \hat{e}_a^T Q_{yy}^{-1} \hat{e}_a > k_\alpha$$

Four representations for left hand side:

1) $$\underline{T}_q = \underline{\hat{e}}_o^T Q_{yy}^{-1} \underline{\hat{e}}_o - \underline{\hat{e}}_a^T Q_{yy}^{-1} \underline{\hat{e}}_a$$

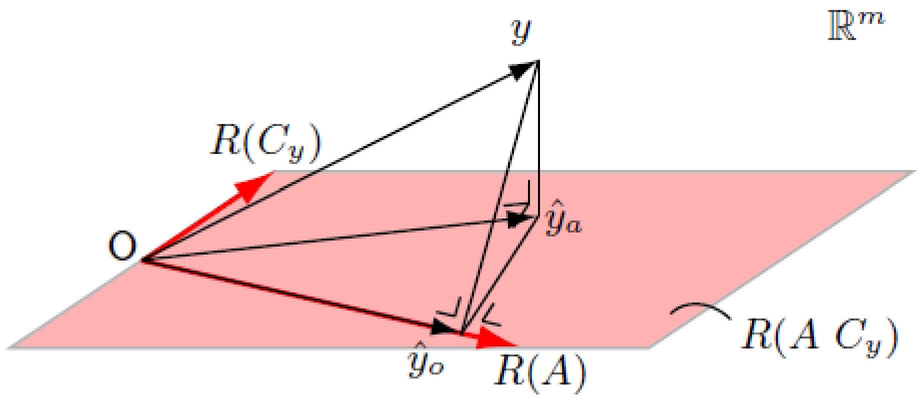
2) $$T_q = (\hat{y}_o - \hat{y}_a)^T Q_{yy}^{-1} (\hat{y}_o - \hat{y}_a)$$

3) $$T_q = \hat{\nabla}^T C_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} C_y \hat{\nabla}$$

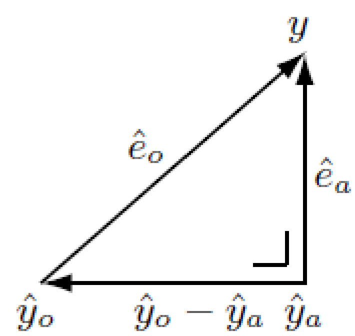
4) $$T_q = \hat{e}_o^T Q_{yy}^{-1} C_y (C_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} C_y)^{-1} C_y^T Q_{yy}^{-1} \hat{e}_o$$

Mathematical/Statistical Background

Geometrical interpretation:



$$\|\hat{e}_o\|^2_{Q_{yy}^{-1}} - \|\hat{e}_a\|^2_{Q_{yy}^{-1}} = \|\hat{y}_o - \hat{y}_a\|^2_{Q_{yy}^{-1}}$$



Mathematical/Statistical Background

If the original observations are normally distributed, one has

$$H_o: \underline{T}_q \sim \chi^2(q,0) \text{ and } H_a: \underline{T}_q \sim \chi^2(q,\lambda)$$

with noncentrality parameter

$$\lambda = \nabla^T Q_{\hat{\nabla}\hat{\nabla}}^{-1} \nabla = \nabla^T C_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} C_y \nabla$$

- This can have a variety of applications in functional model identification.
  - Outliers
  - Offsets
  - Periodic patterns
- Two of well-known applications is the overall model test and the w-test statistics.

Mathematical/Statistical Background

Two extreme cases:

Overall model test: $q = m - n$

$$\underline{T}_{q=m-n} = \hat{\underline{e}}_o^T Q_{yy}^{-1} \hat{\underline{e}}_o \qquad \hat{e}_a = 0$$

$$H_o: \underline{T}_q \sim \chi^2(m-n, 0) \text{ and } H_a: \underline{T}_q \sim \chi^2(m-n, \lambda)$$

W-test: $q = 1$

$$\underline{T}_{q=1} = \frac{(c_y^T Q_{yy}^{-1} \hat{\underline{e}}_o)^2}{c_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} c_y} \qquad H_o: \underline{T}_q \sim \chi^2(1, 0) \text{ and } H_a: \underline{T}_q \sim \chi^2(1, \lambda)$$

Mathematical/Statistical Background

W-test statistics (Outlier detection):

If we apply square root to the previous formula, we have:

$$\underline{w} = \frac{c_y^T Q_{yy}^{-1} \underline{\hat{e}}_o}{\sqrt{c_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} c_y}} \qquad H_o : \ \underline{w} \sim \mathrm{N}(0,1) \text{ and } H_a : \ \underline{w} \sim \mathrm{N}(\nabla w, 1)$$

For a diagonal covariance matrix, and

$$c_{y_i} = (0, \ldots, 1, \ldots, 0)^T$$

we have

$$\underline{w}_i = \frac{\underline{\hat{e}}_i}{\sigma_{\hat{e}_i}} \qquad H_o \ \underline{w}_i \sim \mathrm{N}(0,1) \text{ and } H_a : \underline{w}_i \sim \mathrm{N}(\nabla w, 1)$$

Functional model identification

Identification of deterministic effects in GNSS time series:

H_0: $E(y) = Ax, D(y) = Q_y$

H_a: $E(y) = Ax + A_k x_k\,, D(y) = Q_y$

For a linear regression model with a periodic function we have

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}$$

$$A_k = \begin{bmatrix} \cos \omega_k t_1 & \sin \omega_k t_1 \\ \cos \omega_k t_2 & \sin \omega_k t_2 \\ \vdots & \vdots \\ \cos \omega_k t_m & \sin \omega_k t_m \end{bmatrix}$$

Functional model identification

Periodic functions:

H$_0$: $E(y) = Ax, D(y) = Q_y$

H$_a$: $E(y) = Ax + A_k x_k\,, D(y) = Q_y$

For a linear regression model with a periodic function we have

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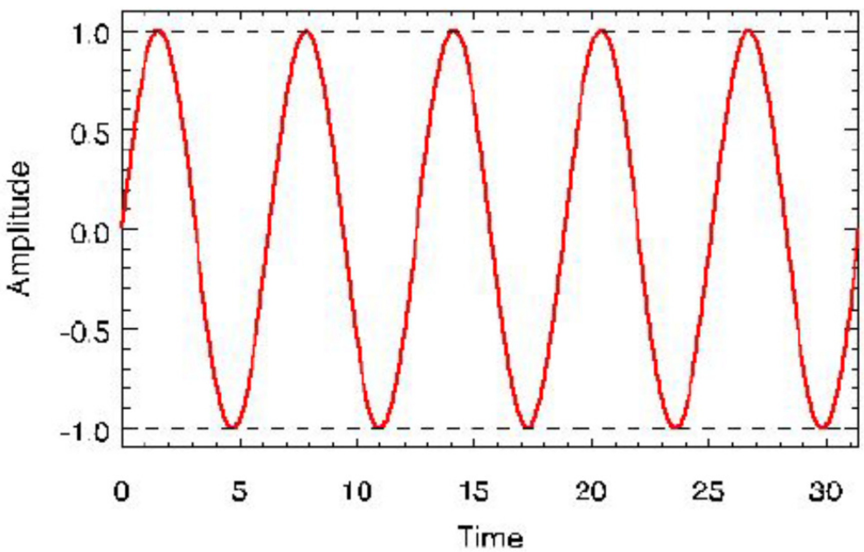
Functional model identification

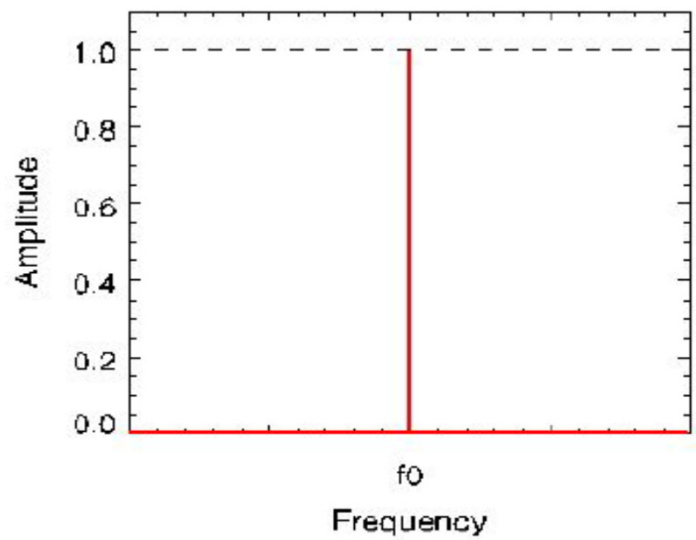
Sinusoidal function:

$$y(t) = Asin(2\pi f_0 t + \phi)$$

Time domain:



Frequency domain:

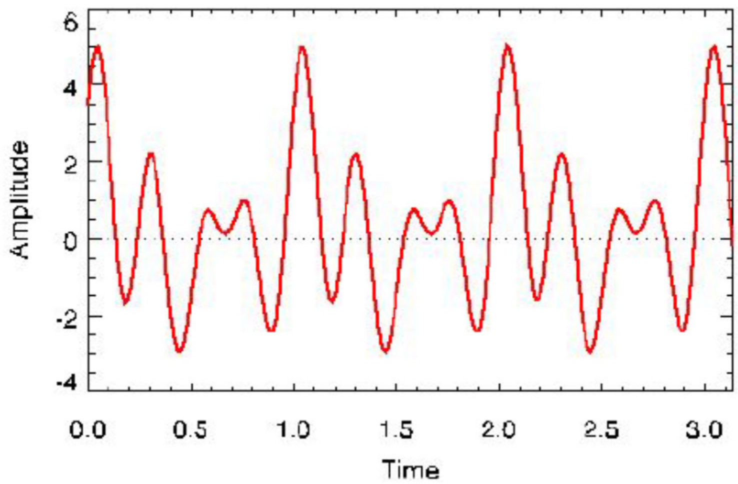


Functional model identification

Complex sinusoidal functions: $y(t) = y(t \pm nT)$, n=1,2,…

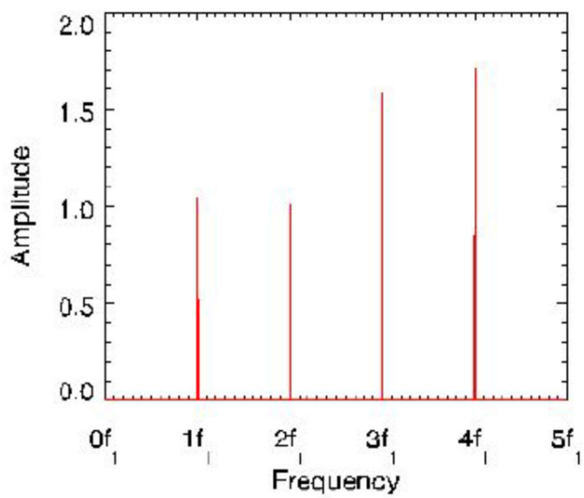
$$\boldsymbol{y}(\boldsymbol{t}) = \frac{\boldsymbol{a_0}}{\boldsymbol{2}} + \sum_{\boldsymbol{n=1}}^{\boldsymbol{4}} a_n \cos(2\pi n f_1 t) + b_n \sin(2\pi n f_1 t)$$

Time domain:



Frequency domain:

(Spectrogram)



Functional model identification

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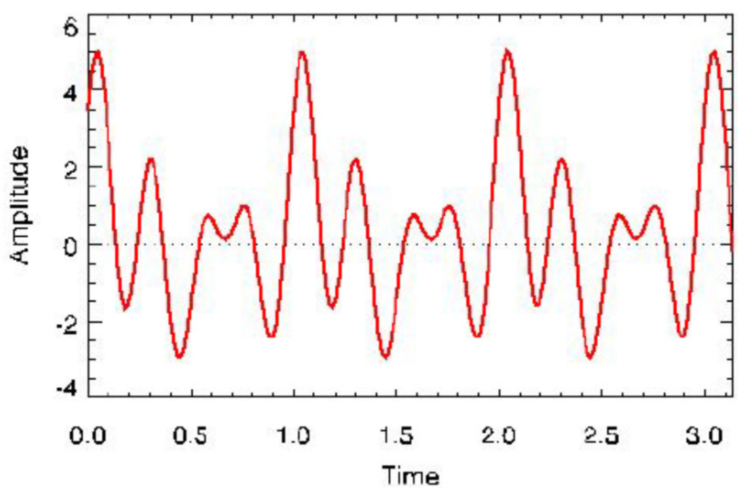
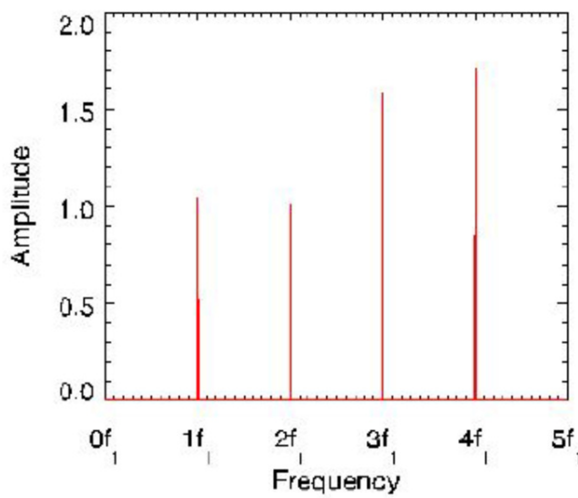
Time domain:



Frequency domain:

(Spectrogram)



Discrete Fourier Transform

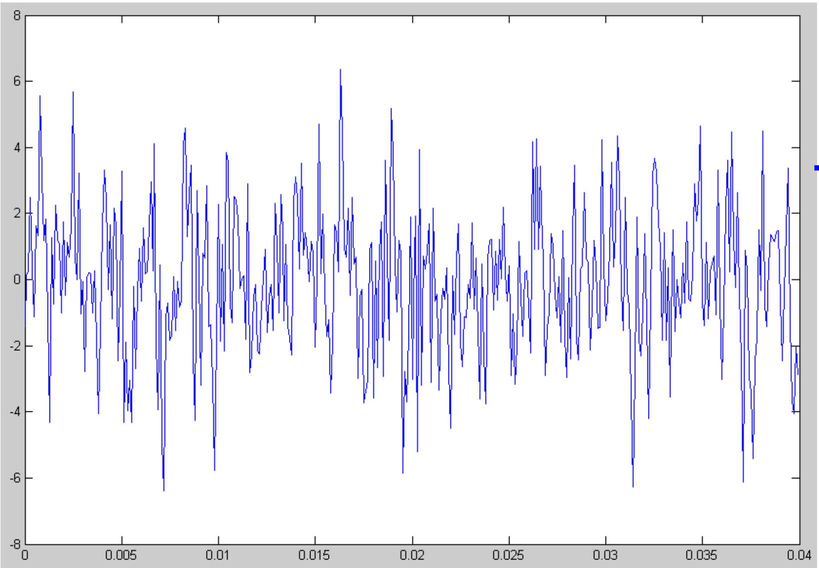
Fourier Transform

Signal Analysis and Processing

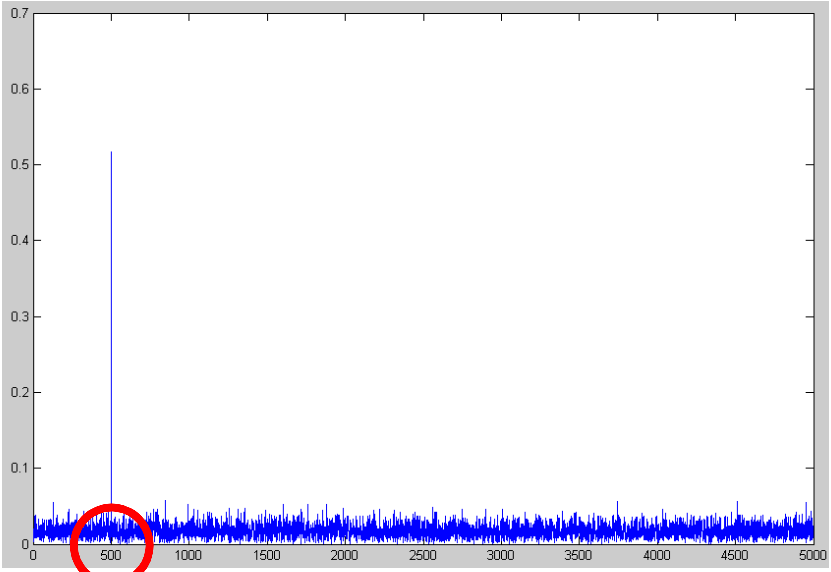
（1）Time Domain Analysis: t-A

$$x(t) = \sin(2\pi \times 500 \times t)$$

（2）Frequency Domain Analysis: f-A



Fourier Transform



$x(t)$ **in time-domain**

$$x(t) = \sin(2\pi \times f \times t) \quad + 2 \times randn(\ )$$

$x(t)$ **in frequency-domain**

In some situation, signal's frequency spectrum can represent its characteristics more clearly.

Fourier Transform

Signal Analysis and Processing:

（1） Time Domain Analysis

（2） Frequency Domain Analysis

Fourier Transform is a bridge from time domain to frequency domain.

Characteristic: continuous—discrete, periodic—nonperiodic.

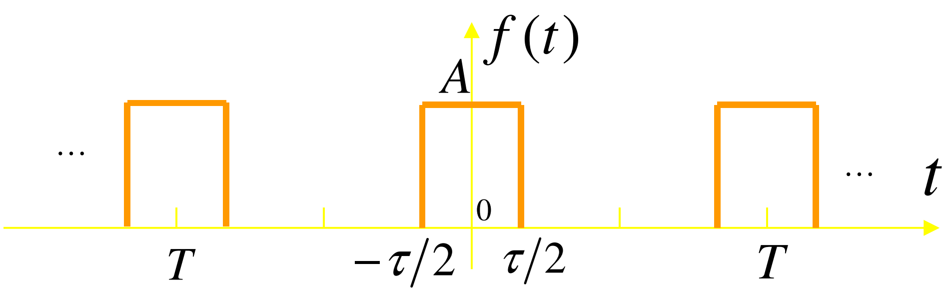
Types:

- Continuous periodic signals
- Continuous nonperiodic signals
- Discrete nonperiodic signals
- Discrete periodic signals

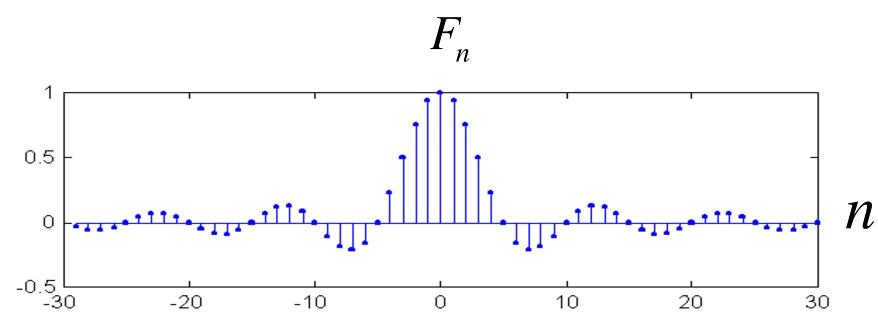
Continuous periodic signal--Fourier Series

It is proved that continuous-time periodic signal can be represented by a Fourier Series corresponding to a sum of harmonically related complex exponential signal. To a periodic function $f(t)$ with period T

$$f(t) = \sum_{n=-\infty}^{+\infty} F_n e^{jn\omega_0 t}, \ \omega_0 = \frac{2\pi}{T} \Leftrightarrow F_n = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-jn\omega_0 t} dt$$



Time-domain



Frequency-domain

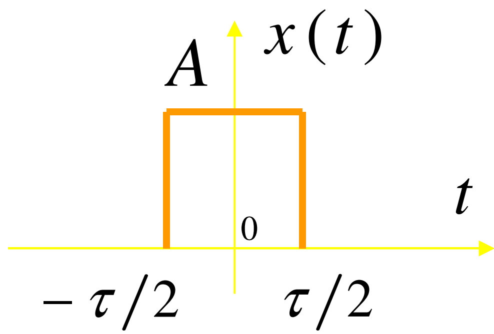
Conclusion: Continuous periodic function—

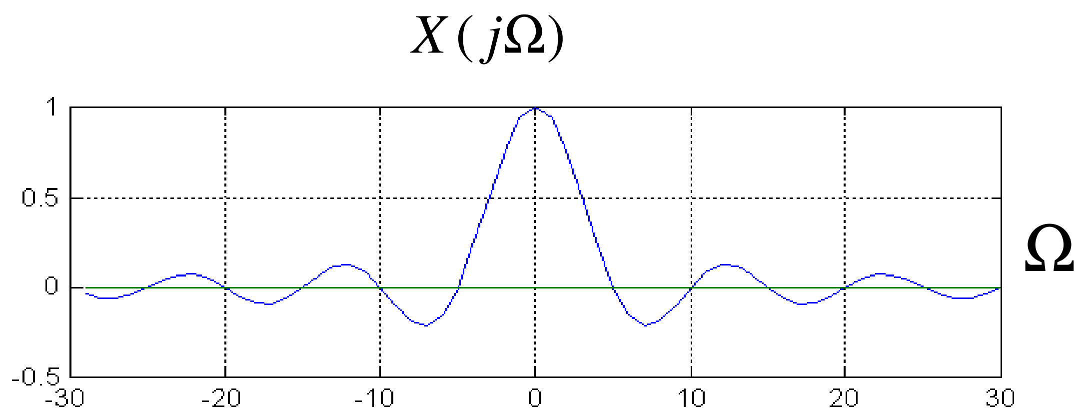
Nonperiodic discrete frequency impulse sequence

Continuous nonperiodic function's Fourier Transform

$$X(j\Omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\Omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\Omega) e^{j\Omega t} d\Omega$$





Time-domain **Frequency-domain**

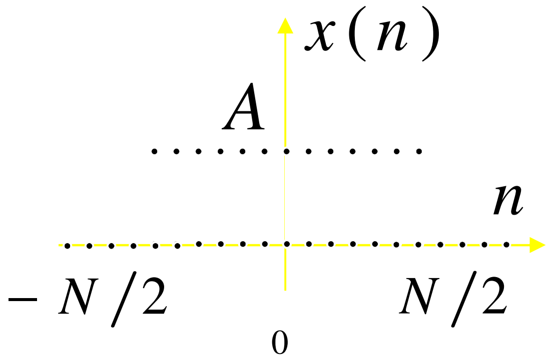
Conclusion : Continuous nonperiodic function—
Nonperiodic continuous function

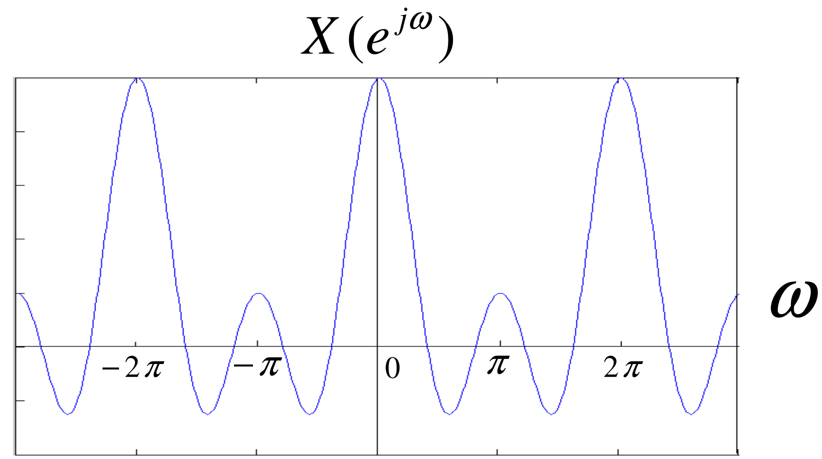
Discrete-time nonperiodic sequence's Fourier Transform

$$\begin{cases} X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x(n) e^{-j\omega n} \\ x(n) = \frac{1}{2\pi}\int_{-\pi}^{+\pi} X(e^{j\omega}) e^{j\omega n} d\omega \end{cases}$$

$$X(e^{j(\omega+2\pi M)}) = \sum_{n=-\infty}^{+\infty} x(n) e^{-j(\omega+2\pi M)n}$$

$$= \sum_{n=-\infty}^{+\infty} x(n) e^{-j\omega n} e^{-j2\pi nM} = \sum_{n=-\infty}^{+\infty} x(n) e^{-j\omega n}$$





Time-domain

Frequency-domain

Conclusion : Discrete nonperiodic function—
Continuous-time periodic function

Conclusion

（1）Sampling in time domain brings periodicity in frequency domain.

（2）Sampling in frequency domain brings periodicity in time domain.

（3）Relationship between frequency domain and time domain

Time domain	Frequency domain	Transform
Continuous periodic	Discrete nonperiodic	Fourier series
Continuous nonperiodic	Continuous nonperiodic	Fourier Transform
Discrete nonperiodic	Continuous periodic	Sequence's Fourier Transform
Discrete periodic	Discrete periodic	Discrete Fourier Series

Periodic ←→ Discrete

Nonperiodic←→Continuous

Basic idea of Discrete Fourier Transform

In practical applications, signal processed has two main characteristics:

(1) Discrete **(2) Finite length**

Similarly, signal's frequency must also have two main characteristics:

(1) Discrete **(2) Finite length**

But nonperiodic sequence's Fourier Transform is a continuous function of ω, and it is a periodic function in ω with a period 2π. So it is not suitable to solve many practical problems.

Idea:

Expand finite-length sequence to periodic sequence, compute its Discrete Fourier Series, so that we can get the discrete spectrum in frequency domain.

Discrete Fourier Series

1) Discrete Fourier Series Transform Pair

Similar with continuous-time periodic signals, a periodic sequence $\tilde{x}(n)$ with period N, can be represented by a Fourier Series corresponding to a sum of harmonically related complex exponential sequences, such as:

$$\tilde{x}(n) = \frac{1}{N}\sum_{k=0}^{N-1}\tilde{X}(k)e^{j\omega_k n} \qquad (**)$$

where $\omega_1 = \frac{2\pi}{N}$, $\tilde{x}(n)$'s fundamental angular frequency

$\omega_k = \frac{2\pi}{N}k$, $\tilde{x}(n)$'s kth harmonic angular frequency

Attention: Fourier Series for discrete-time signal with period N requires only N harmonically related complex exponentials.

$$\because e^{j\frac{2\pi}{N}kn} = e^{j\frac{2\pi}{N}k(n+N)}, \ \text{that is}: \omega_{k+N} = \omega_k$$

$\tilde{X}(k)$ computation

After multiplying both sides of Eq.(**) with $e^{-j\frac{2\pi}{N}kn}$, and summing from $n=0$ *to* N -1, we obtain

$$\sum_{n=0}^{N-1}\tilde{x}(n)e^{-j\frac{2\pi}{N}kn}=\frac{1}{N}\sum_{n=0}^{N-1}\sum_{r=0}^{N-1}\tilde{X}(r)e^{j\frac{2\pi}{N}rn}e^{-j\frac{2\pi}{N}kn}$$

$$=\frac{1}{N}\sum_{n=0}^{N-1}\sum_{r=0}^{N-1}\tilde{X}(r)e^{j\frac{2\pi}{N}(r-k)n}=\sum_{r=0}^{N-1}\tilde{X}(r)\underline{\left(\frac{1}{N}\sum_{n=0}^{N-1}e^{j\frac{2\pi}{N}(r-k)n}\right)}$$

$$\frac{1}{N}\sum_{n=0}^{N-1}e^{j\frac{2\pi}{N}(r-k)n}=\frac{1}{N}\cdot\frac{1-e^{j\frac{2\pi}{N}(r-k)N}}{1-e^{j\frac{2\pi}{N}(r-k)}}=\begin{cases}1 & r=k+mN\\ 0 & \text{otherwise}\end{cases}$$

$$\Rightarrow\tilde{X}(r)=\tilde{X}(k)=\sum_{n=0}^{N-1}\tilde{x}(n)e^{-j\frac{2\pi}{N}kn}$$

Discrete Fourier Series for periodic sequence:

$$\tilde{X}(k) = DFS[\tilde{x}(n)] = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} \tilde{x}(n) W_N^{kn}$$

$$W_N = e^{-j\frac{2\pi}{N}}$$

$$\tilde{x}(n) = IDFS[\tilde{X}(k)] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}(k) e^{j\frac{2\pi}{N}kn} = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{X}(k) W_N^{-kn}$$

Attention:

$$\tilde{X}(k+mN) = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}(k+mN)n} = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}kn} = \tilde{X}(k)$$

so: $\tilde{X}(k)$ is also a periodic sequence with period N.

$\tilde{X}(k)$ and $\tilde{x}(n)$ is a period sequence pair in frequency domain and time domain.

Properties of DFS

Suppose $DFS[\tilde{x}(n)] = \tilde{X}(k),\ \ DFS[\tilde{y}(n)] = \tilde{Y}(k)$

（1） **Linear**

$$\boldsymbol{DFS[a\tilde{x}(n) + b\tilde{y}(n)] = a\tilde{X}(k) + b\tilde{Y}(k)}$$

（2） **Sequence Shift**

$$DFS[\tilde{x}(n+m)] = W_N^{-km}\tilde{X}(k)$$

$$IDFS[\tilde{X}(k+l)] = W_N^{nl}\tilde{x}(n)$$

$$Proof: DFS[\tilde{x}(n+m)] = \sum_{n=0}^{N-1}\tilde{x}(n+m)W_N^{kn}, let\ r = n+m:$$

$$Then = \sum_{r=m}^{N+m-1}\tilde{x}(r)W_N^{k(r-m)} = W_N^{-km}\sum_{r=m}^{N+m-1}\tilde{x}(r)W_N^{kr}$$

$$= W_N^{-km}\sum_{r=0}^{N-1}\tilde{x}(r)W_N^{kr} \Rightarrow DFS[\tilde{x}(n+m)] = W_N^{-km}\tilde{X}(k)$$

Properties of DFS

Suppose $DFS[\tilde{x}(n)] = \tilde{X}(k)$, $DFS[\tilde{y}(n)] = \tilde{Y}(k)$

(3) Periodic Convolution

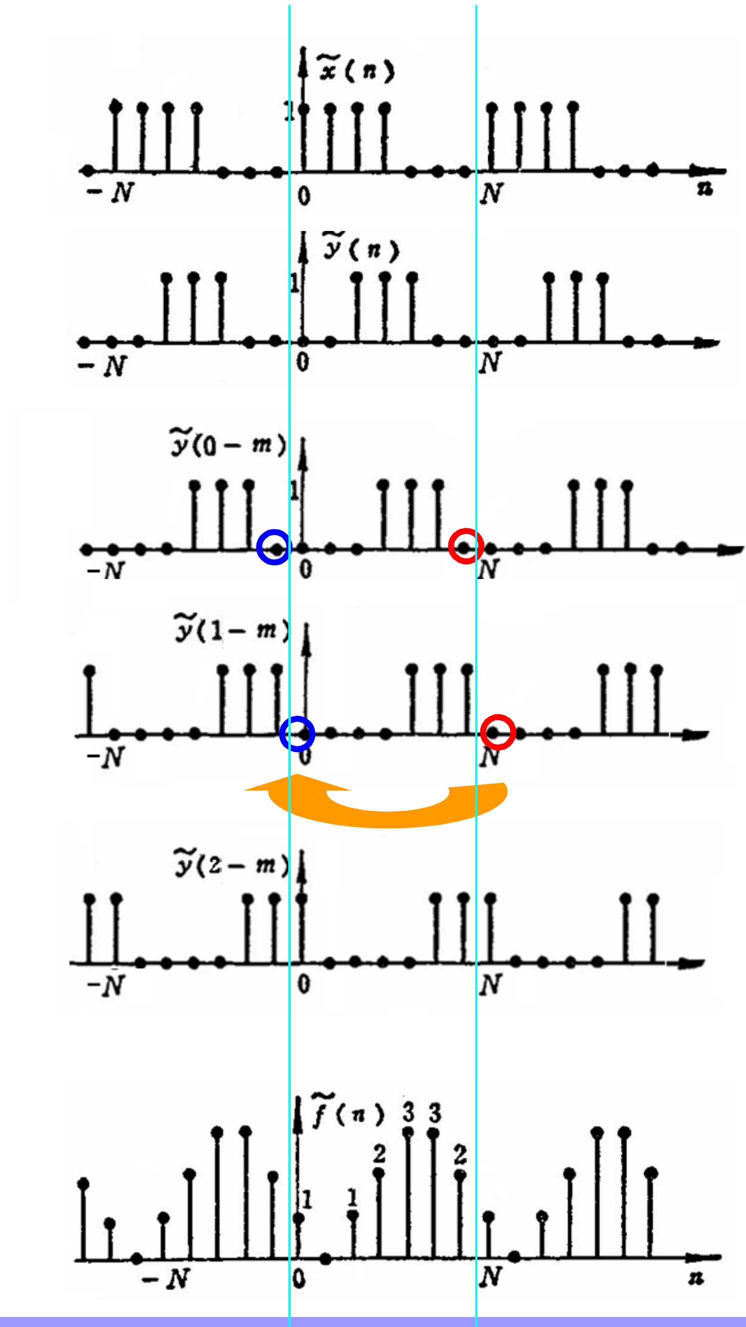
If $\tilde{F}(k) = \tilde{X}(k)\tilde{Y}(k)$, then:

$$\tilde{f}(n) = IDFS[\tilde{F}(k)] = \underline{\sum_{m=0}^{N-1} \tilde{x}(m)\tilde{y}(n-m)} = \underline{\sum_{m=0}^{N-1} \tilde{y}(m)\tilde{x}(n-m)}$$

Periodic convolution

Compared with linear convolution, periodic convolution's main difference is:

The sum is over the finite interval m=0~N-1.



Periodic convolution

$$\tilde{f}(n) = \sum_{m=0}^{N-1} \tilde{x}(m)\,\tilde{y}(n-m)$$

$$\tilde{f}(0) = \sum_{m=0}^{N-1} \tilde{x}(m)\,\tilde{y}(0-m) = 1$$

$$\tilde{f}(1) = \sum_{m=0}^{N-1} \tilde{x}(m)\,\tilde{y}(1-m) = 0$$

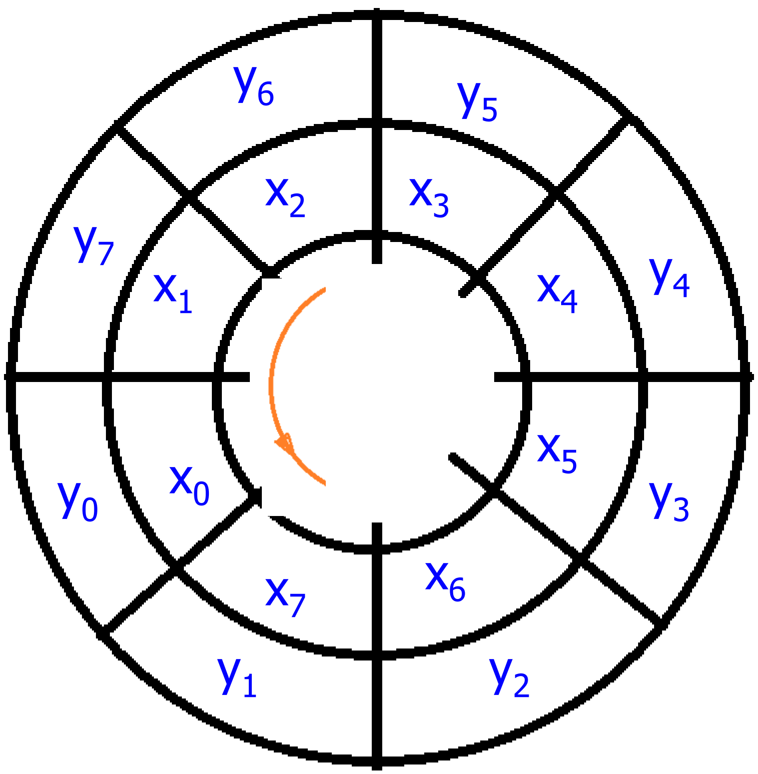
$$\tilde{f}(2) = \sum_{m=0}^{N-1} \tilde{x}(m)\,\tilde{y}(2-m) = 1$$

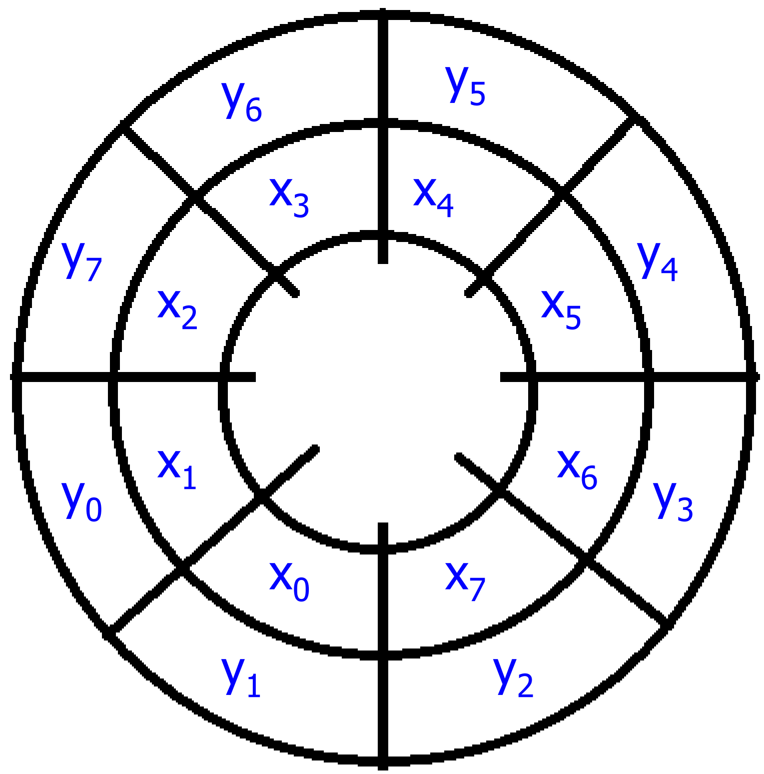
Periodic convolution

$x(n)=[x_0\ x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7]$

$y(n)=[y_0\ y_1\ y_2\ y_3\ y_4\ y_5\ y_6\ y_7]$

$x(n) \circledast y(n)$





Symmetry:

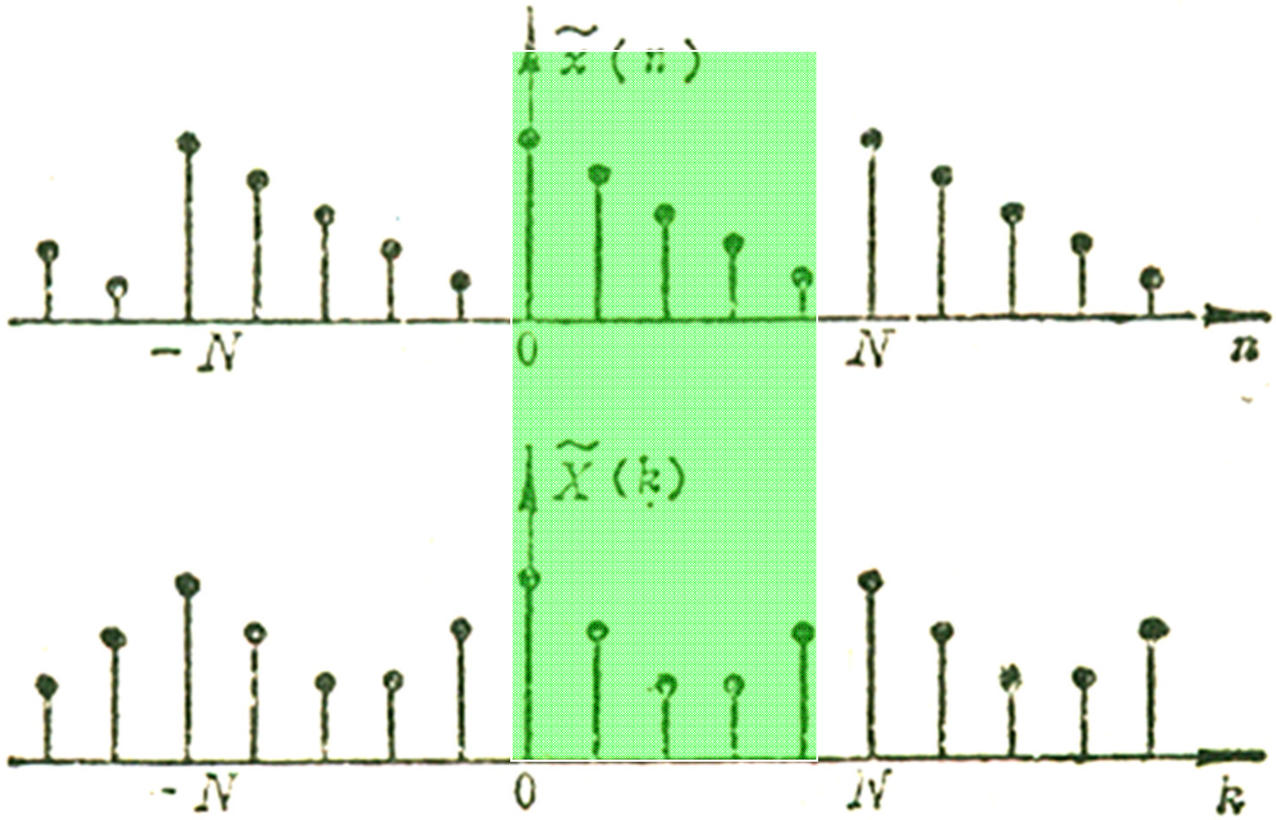
If $\tilde{f}(n) = \tilde{x}(n)\tilde{y}(n)$, then:

$$\tilde{F}(k) = DFS[\tilde{f}(n)] = \frac{1}{N}\sum_{l=0}^{N-1}\tilde{X}(l)\tilde{Y}(k-l) = \frac{1}{N}\sum_{l=0}^{N-1}\tilde{Y}(l)\tilde{X}(k-l)$$

Multiplication of periodic sequence in time-domain corresponds to convolution of periodic sequence in frequency domain.

Discrete Fourier Transform-DFT

Periodic sequence and its DFS



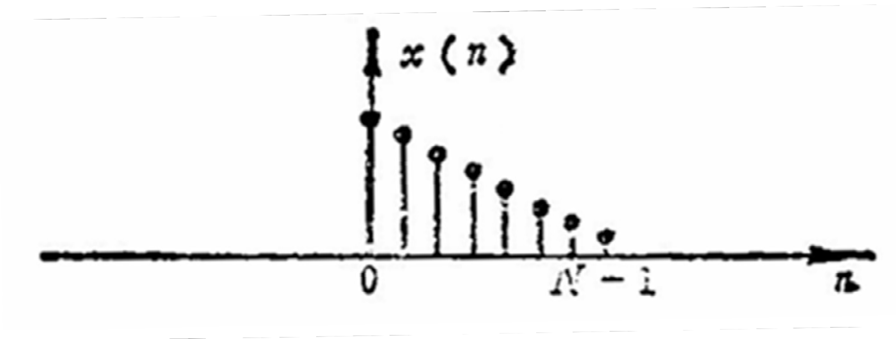
Discrete Fourier Transform-DFT

Relationship between periodic sequence and finite-length sequence

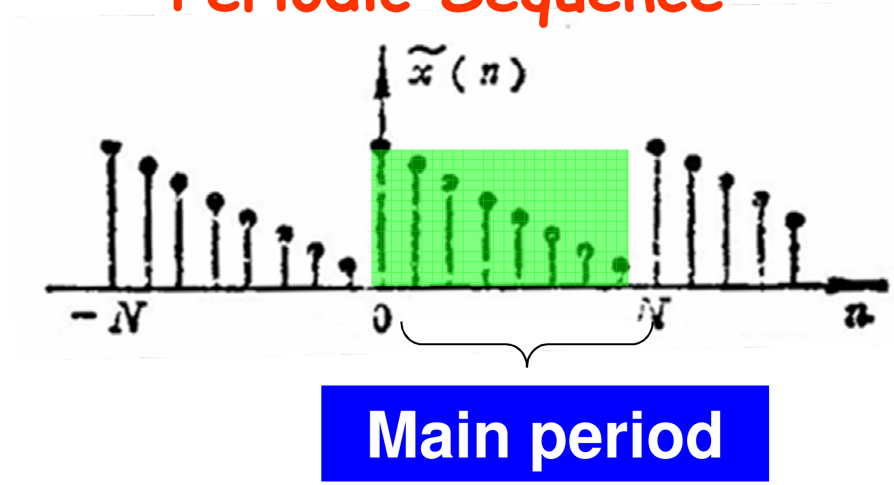
Periodic sequence can be seen as periodically copies of finite-length sequence.

Finite-length sequence can be seen as extracting one period from periodic sequence.

Finite-duration Sequence



Periodic Sequence



Discrete Fourier Transform

- The DFS pair

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn} \qquad \tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$

- The equations involve only on period so we can write

$$X[k] = \begin{cases} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn} & 0 \le k \le N-1 \\ 0 & \text{else} \end{cases}$$

$$x[n] = \begin{cases} \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn} & 0 \le k \le N-1 \\ 0 & \text{else} \end{cases}$$

- The Discrete Fourier Transform

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn} \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)kn}$$

- The DFT pair can also be written as

$$X[k] \xleftrightarrow{\text{DFT}} x[n]$$

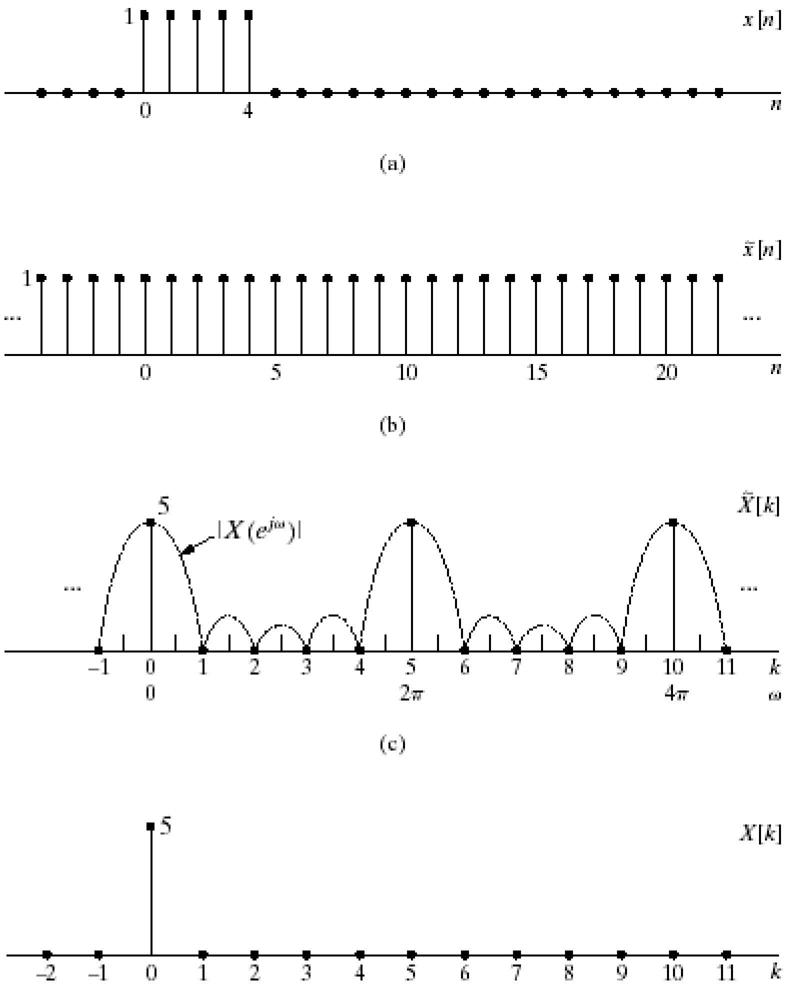
Example

- The DFT of a rectangular pulse
- x[n] is of length 5
- We can consider x[n] of any length greater than 5
- Let's pick N=5
- Calculate the DFS of the periodic form of x[n]

$$\tilde{X}[k] = \sum_{n=0}^{4} e^{-j(2\pi k/5)n}$$

$$= \frac{1 - e^{-j2\pi k}}{1 - e^{-j(2\pi k/5)}}$$

$$= \begin{cases} 5 & k = 0, \pm 5, \pm 10, \ldots \\ 0 & \text{else} \end{cases}$$



Why Using DFT for Linear Convolution?

- FFT (Fast Fourier Transform) exists.
- But, we have to ensure that circular convolving nature of DFT gives the linear convolving result.

The Procedure

1. Let $x_1(n)$ and $x_2(n)$ be two sequences of length L and P, respectively.
2. Compute N-point ($N = ?$) DFTs $X_1(k)$ and $X_2(k)$.
3. Let $X_3(k) = X_1(k)\, X_2(k)$, $0 \le k \le N-1$.
4. Let $x_3(n) = \text{DFT}^{-1}[X_3(k)] = x_1(n) \circledast x_2(n)$.

How to interpret DFT?

Discrete Fourier Transform

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j(2\pi/N)kn}$$

DFT in matrix notation

$$\begin{pmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{pmatrix} = \begin{pmatrix} W_N^0 & W_N^0 & & W_N^0 \\ W_N^0 & W_N^1 & \cdots & W_N^{N-1} \\ \vdots & & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & \cdots & W_N^{(N-1)(N-1)} \end{pmatrix} \begin{pmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{pmatrix}$$

where

$$W_N^{kn} = \frac{1}{N} e^{j(\frac{2\pi}{N})kn} = \frac{1}{N}\cos(\frac{2\pi}{N})kn + j\frac{1}{N}\sin(\frac{2\pi}{N})kn$$

How to interpret DFT?

Discrete Fourier Transform

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j(2\pi/N)kn}$$

For k=1, one has

$$\boldsymbol{x(1) = W_N^0 X(0) + W_N^1 X(1) + \ldots + W_N^{N-1} X(N-1)}$$

which is worked out as

$$x(1) = \frac{1}{N}[\cos(0)+j\sin(0)]X(0) + \frac{1}{N}[\cos(\frac{2\pi}{N})+j\sin(\frac{2\pi}{N})]X(1) + \ldots + \frac{1}{N}[\cos(\frac{2\pi}{N})(N-1)+j\sin(\frac{2\pi}{N})(N-1)]X(N-1)$$

How to interpret DFT?

which simplifies as

$$
\begin{aligned}
x(1) = {} & \frac{1}{N}[\cos(0)+j\ \sin(0)]X(0) + \frac{1}{N}[\cos(\frac{2\pi}{N})(N/2)+j\ \sin(\frac{2\pi}{N})(N/2)]X(N/2) + \\
& \frac{1}{N}[\cos(\frac{2\pi}{N})+j\ \sin(\frac{2\pi}{N})]X(1) + \frac{1}{N}[\cos(\frac{2\pi}{N})(N/2+1)+j\ \sin(\frac{2\pi}{N})(N/2+1)]X(N/2+1) \\
& \vdots \\
& + \frac{1}{N}[\cos(\frac{2\pi}{N})(N/2-1)+j\ \sin(\frac{2\pi}{N})(N/2-1)]x(N/2-1) + \frac{1}{N}[\cos(\frac{2\pi}{N})(N-1)+j\ \sin(\frac{2\pi}{N})(N-1)]x(N-1)
\end{aligned}
$$

which can be further worked out as

$$
x(1) = a_0\cos(0)+a_{N/2}\cos(\pi)+a_1\cos(\frac{2\pi}{N})+b_1\sin(\frac{2\pi}{N})+\ldots+a_{N/2-1}\cos(\frac{2\pi}{N})(N/2-1)+b_{N/2-1}\ \sin(\frac{2\pi}{N})(N/2-1)
$$

How to interpret DFT?

N number of observation equations are

$$x(0)=a_0\cos(0)+a_{N/2}\cos(\mathbf{0})+a_1\cos(\mathbf{0})+b_1\sin(\mathbf{0})+\ldots+a_{N/2-1}\cos(\mathbf{0})(N/2\text{-}1)+b_{N/2-1}\ \sin(\mathbf{0})(N/2\text{-}1)$$

$$x(1)=a_0\cos(0)+a_{N/2}\cos(\boldsymbol{\pi})+a_1\cos(\frac{2\pi}{N})+b_1\sin(\frac{2\pi}{N})+\ldots+a_{N/2-1}\cos(\frac{2\pi}{N})(N/2\text{-}1)+b_{N/2-1}\ \sin(\frac{2\pi}{N})(N/2\text{-}1)$$

$$\vdots$$

$$x(N\text{-}1)=a_0\cos(0)+a_{N/2}\cos(\boldsymbol{\pi}(N\text{-}1))+a_1\cos(\frac{2\pi(N\text{-}1)}{N})+b_1\sin(\frac{2\pi(N\text{-}1)}{N})+\ldots+a_{N/2-1}\cos(\frac{2\pi(N\text{-}1)}{N})(N/2\text{-}1)+b_{N/2-1}\ \sin(\frac{2\pi(N\text{-}1)}{N})(N/2\text{-}1)$$

N observations – N unknowns

How to interpret DFT?

Solution to the observation equations is

$$a_n = \frac{\sum_{k=0}^{N-1} x_k \cos(\frac{2\pi}{N})kn}{\sum_{k=0}^{N-1} \cos(\frac{2\pi}{N})kn \times \cos(\frac{2\pi}{N})kn}$$

$$b_n = \frac{\sum_{k=0}^{N-1} x_k \sin(\frac{2\pi}{N})kn}{\sum_{k=0}^{N-1} \sin(\frac{2\pi}{N})kn \times \sin(\frac{2\pi}{N})kn}$$